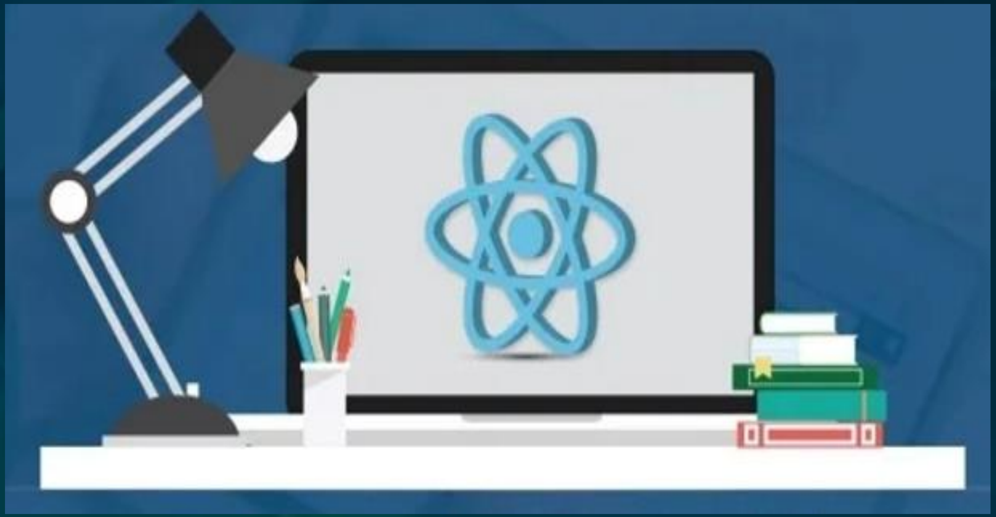
ReactJS

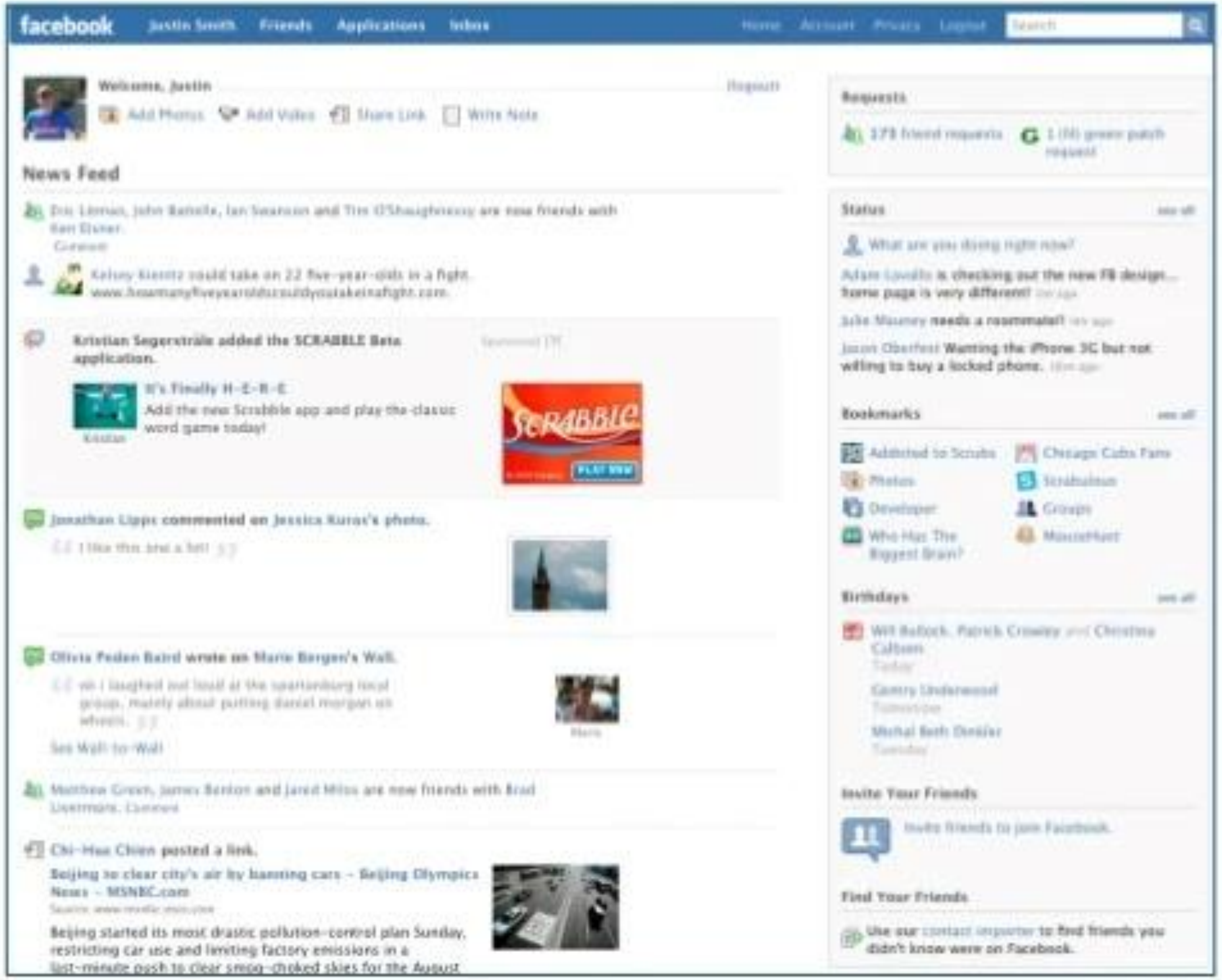
INTRODUCTION



Why ReactJS?

Why do we need ReactJS?

Why ReactJS



Remember the Facebook newsfeeds before 2009, when each time you had to refresh the page for new updates ??

Why ReactJS

facebook
Initial Data
Real Time Data
User Input
Dispatcher
Store
View
Refresh

Why ReactJS

Drawbacks Of Traditional Data Flow

- Uses DOM (Document Object Model)
- More memory consumption
- Slow performance

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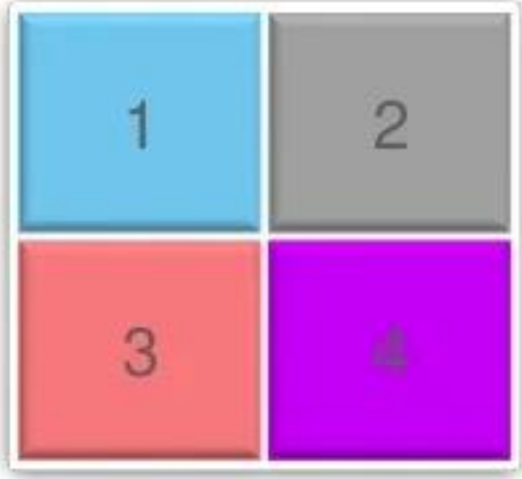
React JS



React JS

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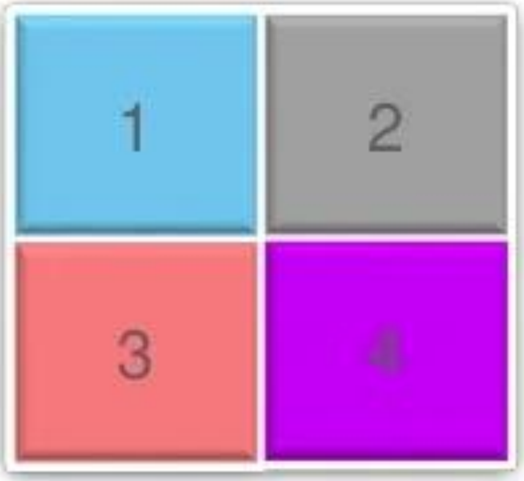

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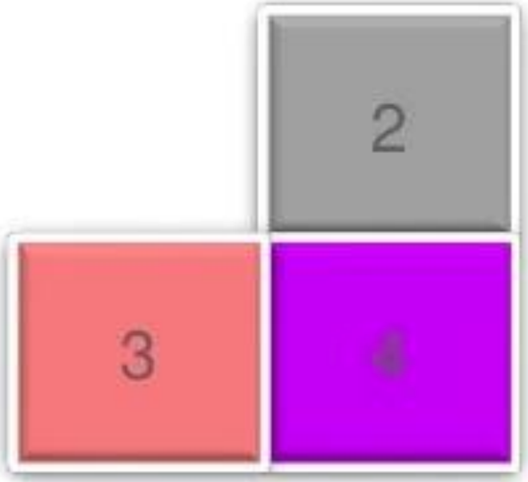

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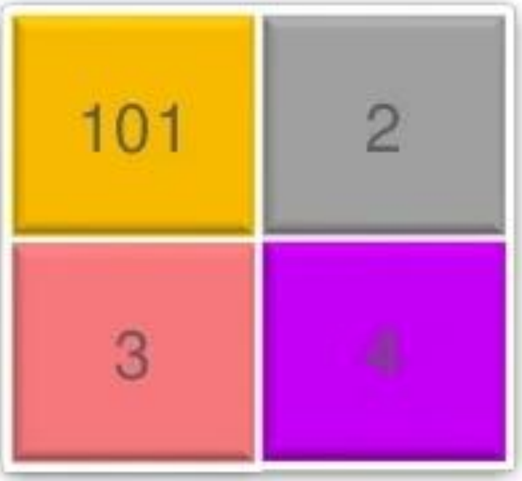
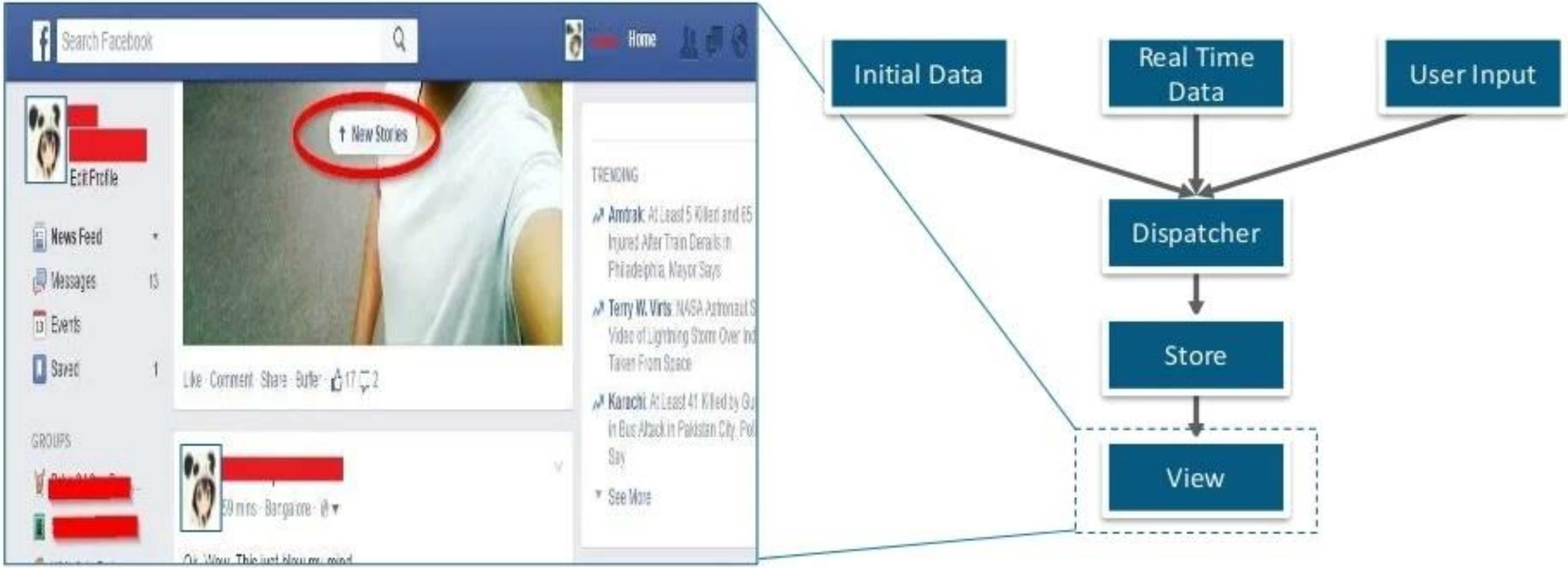


Why ReactJS?

Facebook after implementation of ReactJS became more user friendly. Each time new stories are added, a ↑ New Stories notification appears. Clicking which will automatically refresh the news feeds section.

Why ReactJS?


Search Facebook
Home
Edit Profile
New Stories
News Feed
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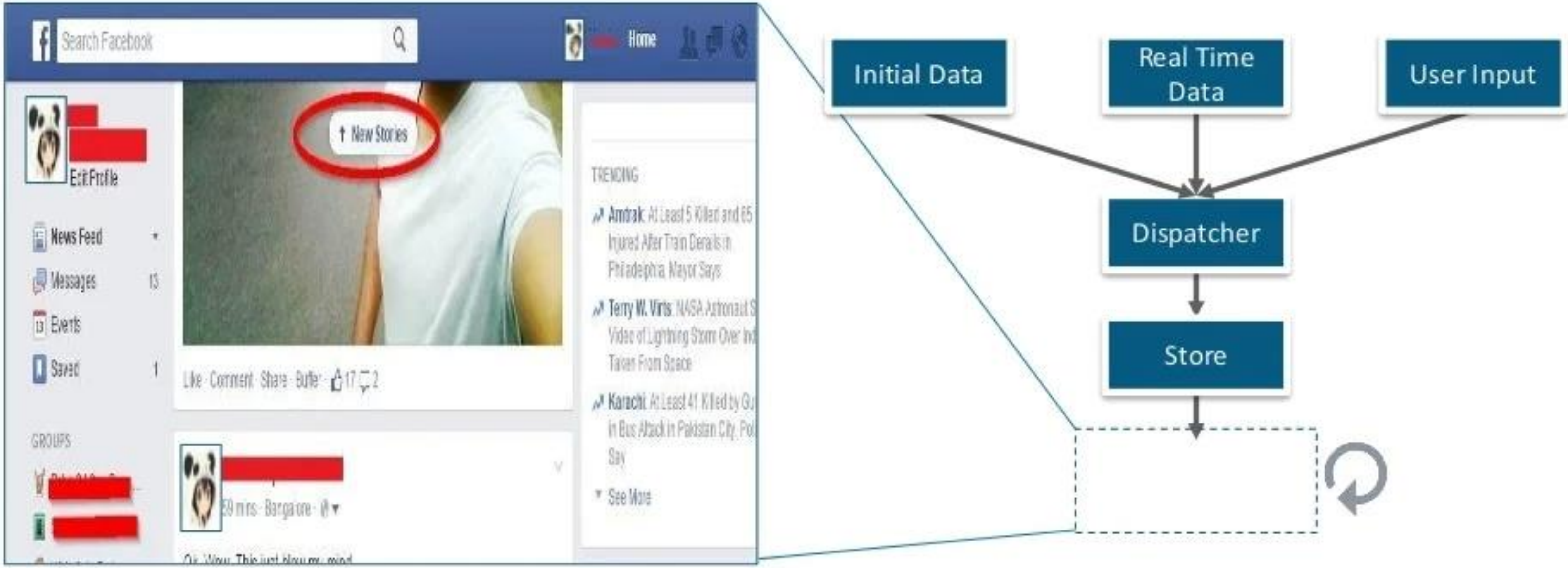
Why ReactJS?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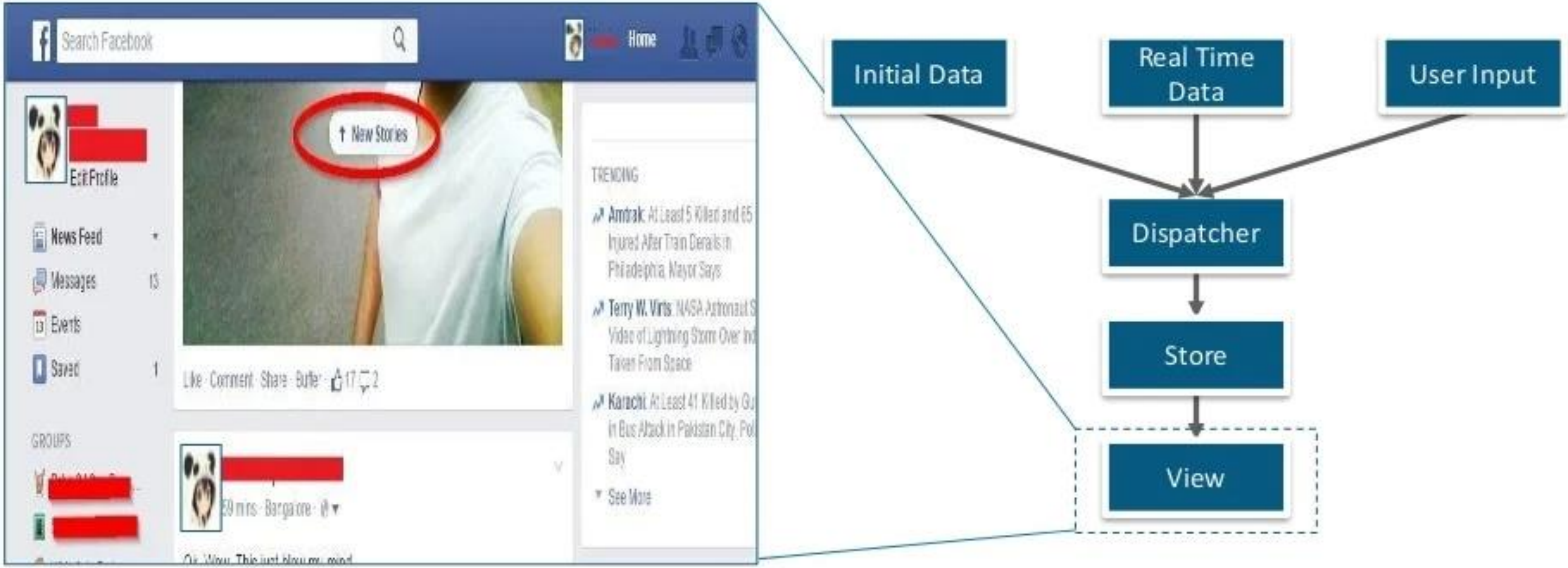
Why ReactJS?



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What Is ReactJS?

ReactJS is an open source JavaScript library used to develop User Interfaces.

- Jordan Walke, a software engineer at Facebook, founded the library in 2011, giving it life.
- Later in 2012, Instagram picks it up and incorporates it into their platform.
- In 2013, ReactJS was released as an open-source project.
- This led to the creation of React Native, a framework fo building native mobile applications using ReactJS.
- In 2015, ReactJS introduced the concept of components, which made it easier to build complex user interfaces.
- This was followed by the introduction of the Virtual DOM, which improved performance and made it possible to create dynamic web applications with ease

What is React?

- React is a JavaScript framework
- Used for front end web development
- Think of jQuery, but more structured
- Created and used by Facebook
- Famous for implementing a Virtual DOM
- React is used to build single-page applications(SPAs).

- React is a JavaScript library that is only responsible for the **view layer** of your application.
- This means it is only responsible for rendering your User Interface (such as the text, text boxes, buttons, etc.) as well as **updating** the UI whenever it changes.

Timeline of front-end JavaScript frameworks

- jQuery* (2006)
- AngularJS (2010)
- React (2013)
- Vue (2014)
- Angular (2014)

* jQuery is more often considered a **library** than a **framework**

React is NOT a framework

React is a library and not a framework.

The difference between a library and a framework is that a library only helps you in one aspect.

In contrast, a framework helps you in many aspects. Let's take an example:

React is a library because it only takes care of your UI.

Angular, on the other hand, is a framework because it handles much more than the UI (It handles Dependency Injection, CSS encapsulation, etc.)

Common tasks in front-end development

App state	Data definition, organization, and storage
User actions	Event handlers respond to user actions
Templates	Design and render HTML templates
Routing	Resolve URLs
Data fetching	Interact with server(s) through APIs and AJAX

Why ReactJs ?

- **Fast** - Feel quick and responsive through the Apps made in React can handle complex updates.
- **Modular** - Allow you to write many smaller, reusable files instead of writing large, dense files of code. The modularity of React is an attractive solution for JavaScript's visibility issues.
- **Scalable** - React performs best in the case of large programs that display a lot of data changes.
- **Flexible** - React approaches differently by breaking them into components while building user interfaces. This is incredibly important in large applications.
- **Popular** - ReactJS gives better performance than other JavaScript languages due to t's implementation of a virtual DOM.
- **Easy to learn** - Since it requires minimal understanding of HTML and JavaScript, the learning curve is low.

Why ReactJs ?

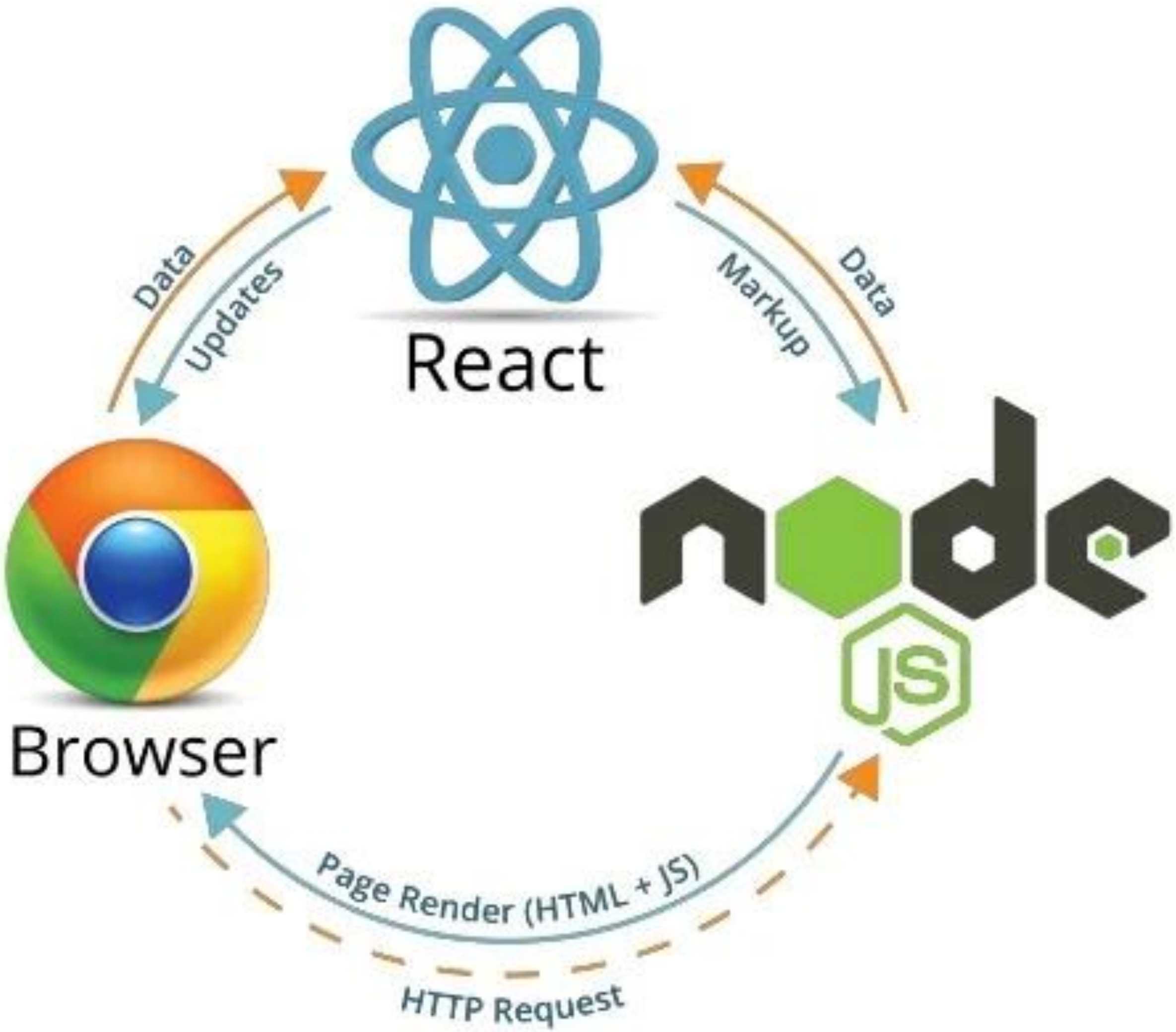
- **Server-side rendering and SEO friendly** - ReactJS websites are famous for their server-side rendering feature. It makes apps faster and much better for search engine ranking in comparison to products with client-side rendering. React even produces more opportunities for website SEO and can occupy higher positions on the search result's page.
- **Reusable UI components** - React improves development and debugging processes.
- **Community** - The number of tools and extensions available for ReactJS developers is tremendous. Along with impressive out-of-box functionalities, more opportunities emerge once you discover how giant the React galaxy is. React has a vibrant community and is supported by Facebook. Hence, it's a reliable tool for website development.

Aspects Of ReactJS?

Virtual DOM

Data Binding

Serverside Rendering



ReactJS Features

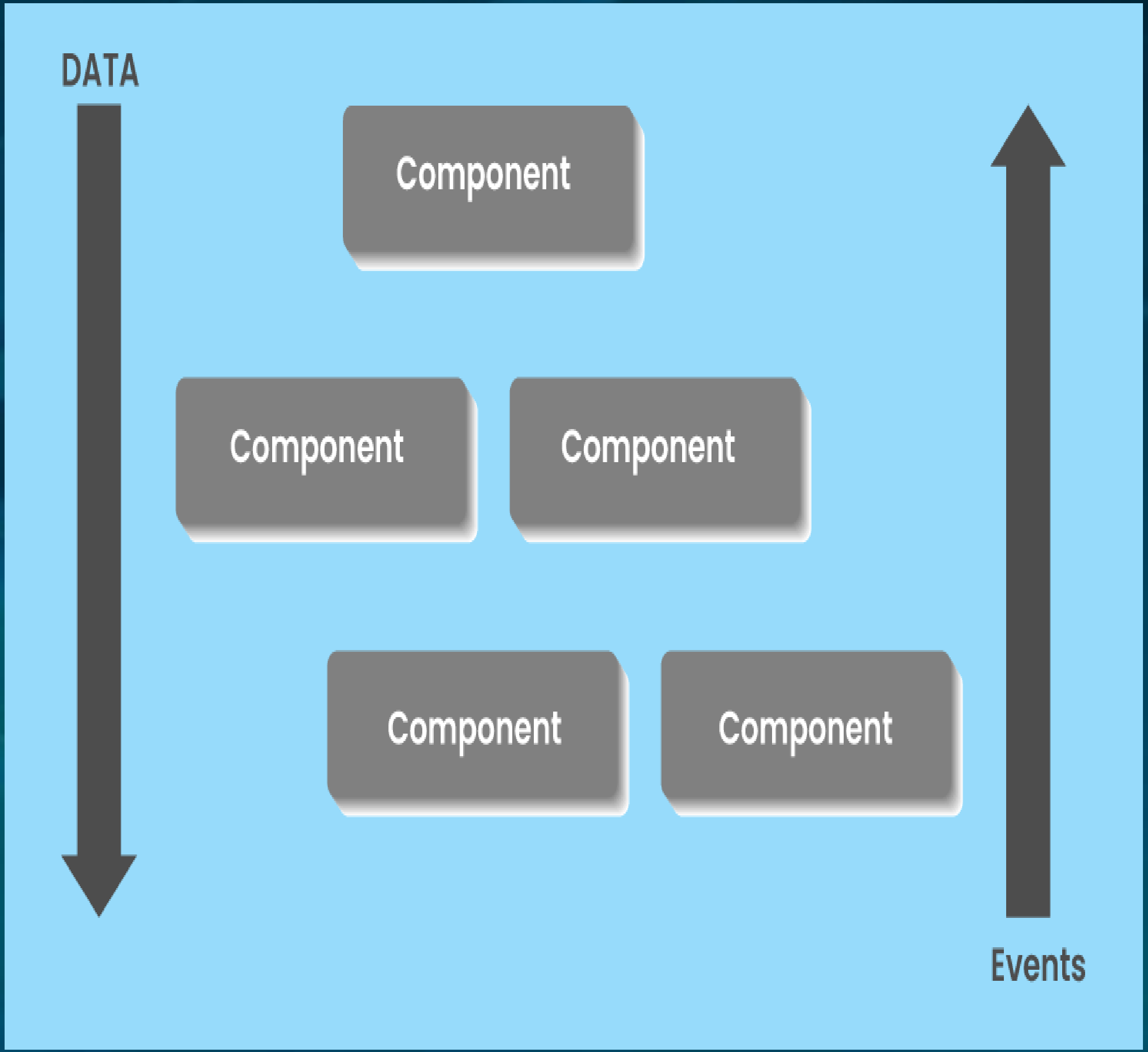
JSX - JavaScript Syntax Extension

- JSX is a preferable choice for many web developers.
- It isn't necessary to use JSX in React development, but there is a massive difference between writing react.js documents in JSX and JavaScript.
- JSX is a syntax extension to JavaScript. By using that, we can write HTML structures in the same file that contains JavaScript code.

ReactJS Features

Unidirectional Data Flow and Flux

- React.js is designed so that it will only support data that is flowing downstream, in one direction.
- If the data has to flow in another direction, you will need additional features.
- React contains a set of immutable values passed to the component rendered as properties in HTML tags.
- The components cannot modify any properties directly but support a call back function to do modifications.



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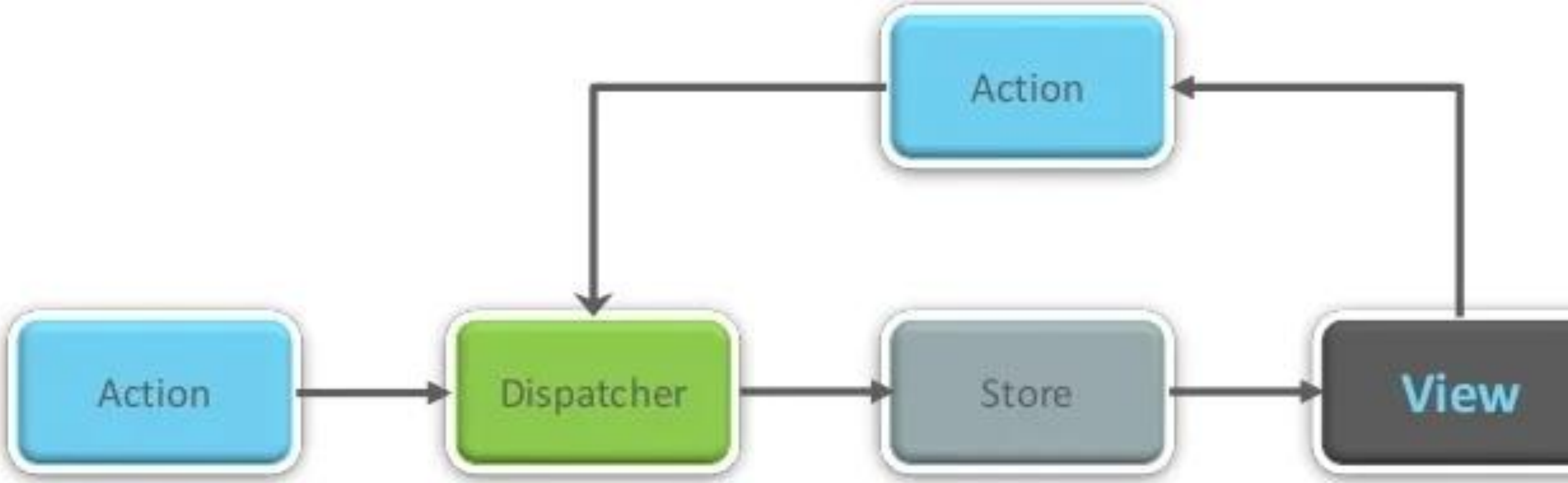


One Way Data Binding

ReactJS Features

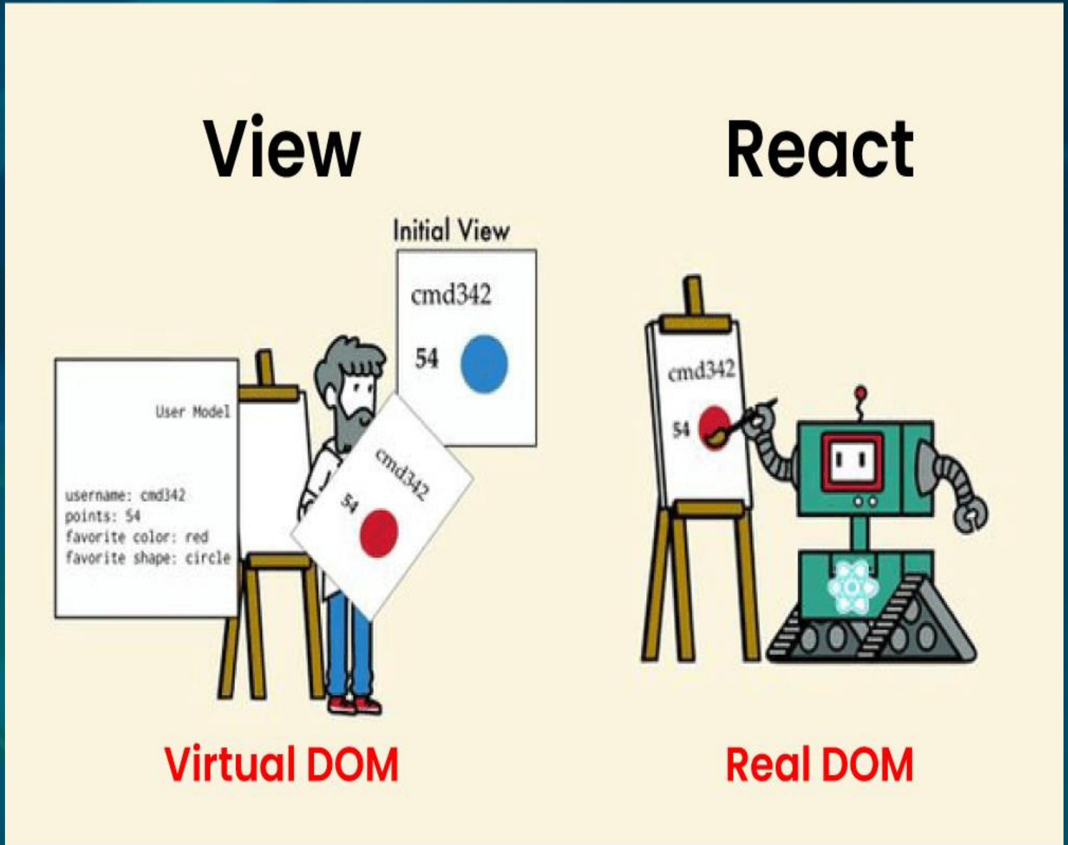
Virtual Document Object Model (VDOM)

- React contains a lightweight representation of real DOM in the memory called Virtual DOM.
- Manipulating real DOM is much slower compared to VDOM as nothing gets drawn on the screen.
- When any object's state changes, VDOM modifies only that object in real DOM instead of updating whole objects.

ReactJS Features

Virtual Document Object Model (VDOM)

- That makes things move fast, particularly compared with other front-end technologies that have to update each object even if only a single object changes in the web application.



ReactJS Features

Extensions

- React supports various extensions for application architecture.
- It supports server-side rendering, Flux, and Redux extensively in web app development.
- React Native is a popular framework developed from React for creating cross-compatible mobile apps.

ReactJS Features

Debugging

- Testing React apps is easy due to large community support.
- Even Facebook provides a small browser extension that makes React debugging easier and faster.

Advantages of ReactJS

Application's performance is increased

Used on Client as well as Server Side

Readability is improved

Easily used with other frameworks

ReactJS Advantages

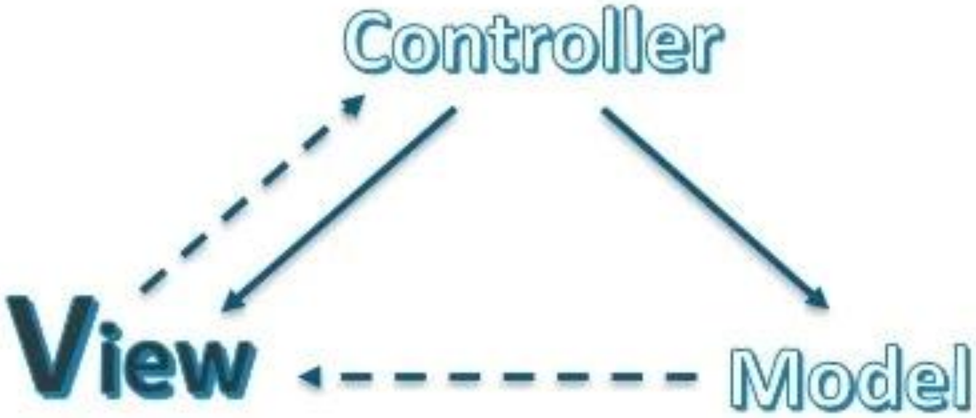
- React.js creates its own virtual DOM. This will improve apps performance since JavaScript virtual DOM is faster than the regular DOM.
- ReactJS helps to create awesome UI.
- ReactJS is SEO-friendly.
- Component and Data patterns improve readability which helps to maintain larger apps.
- React can be used with other frameworks.
- React simplifies the complete process of the scripting environment.

ReactJS Advantages

- It facilitates advanced maintenance and increases productivity.
- Ensures faster rendering
- The best thing about React is the provision of a script for mobile app development.
- A strong community backs ReactJS.
- React JS comes with a helpful developer toolset
- Both startups and fortune 500 companies use React.

What Is ReactJS?

ReactJS is concerned with the components that utilizes the expressiveness of JavaScript with a HTML – like template syntax.



It is basically the View in MVC (Model-**View**-Controller)

Fundamentals of React

- JavaScript and HTML in the same file (JSX)
- Embrace functional programming
- Components everywhere

JavaScript and HTML *in the same file*

HTML CSS JS

JSX CSS or JSS

Traditional approach

React approach

JSX: the React programming language

```
const first = "Aaron";
const last  = "Smith";

const name = <span>{first} {last}</span>;


const list = (
  <ul>
    <li>Dr. David Stotts</li>
    <li>{name}</li>
  </ul>
);
```

```
const listWithTitle = (
  <>
    <h1>COMP 523</h1>
    <ul>
      <li>Dr. David Stotts</li>
      <li>{name}</li>
    </ul>
  </>
);
```

"React is just JavaScript"

Building Components of React

- **Components are the heart and soul of React.**
- **Components (like JavaScript functions) let you split the UI into independent, reusable pieces and think about each piece in isolation.**
- **Components are building blocks of any React application.**
- **Every component has its structures, APIs, and methods.**

ReactJS Components

In React, there are two types of components, namely

Stateless Functional Component

Stateful Class Component

Functional Components

- These components have no state of their own and contain only a return statement.
- They are simply Javascript functions that may or may not receive data as parameters.
- Stateless functional components may derive data from other components as properties (props).
- An example of representing functional component is shown below:

```
function WelcomeMessage(props) {
     return <h1>Welcome to the , {props.name}</h1>;
}
```

Class Components

- These components are more complex than functional components.
- They can manage their state and to return JSX on the screen have a separate render method.
- You can pass data from one class to other class components.
- An example of representing class component is shown below:

```
class MyComponent extends React.Component {
      render() {
      return ( <div>This is the main component.</div> );
} }
```

ReactJs Installation

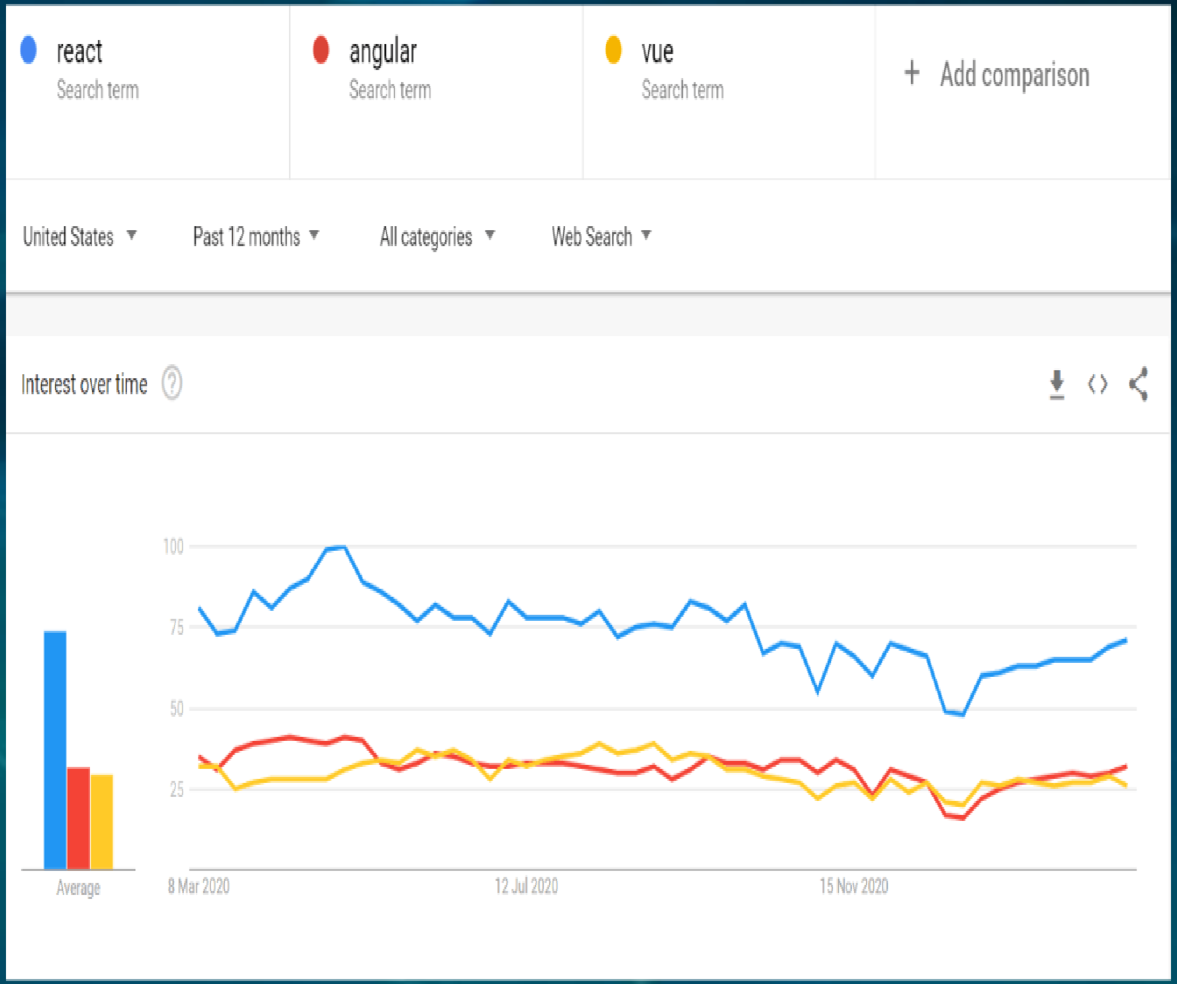
- Step 1: Install Node.js installer for windows from official Web Site.
- Here install the LTS version (the one present on the left).
- Once downloaded open NodeJS without disturbing other settings, click on the Next button until it's completely installed.
- Open command prompt to check whether it is completely installed or not type the command –>
- node –v
- Step 2 : Create a new project folder: mkdir ReactProjects and enter that directory: cd ReactProjects

ReactJs Installation

- Step 3 : Install React using create-react-app, a tool that installs all of the dependencies to build and run a full React.js application.
- npx create-react-app my-app
- npx is the package runner used by npm to execute packages in place of a global install.
- This will first ask for your permission to temporarily install create-react-app and it's associated packages.
- Once completed, change directories into your new app ("my-app" or whatever you've chosen to call it): cd my-app.
- Step 4: Start your new React app: npm start

Conclusion

- React has been the most used web framework by developers worldwide over the years.
- Most of the businesses are inclining towards React because of the simplicity it offers.
- The popularity of React has been consistently gaining, and the Google Trends chart also shows the speed with which it is rising consistently with adding new features over time.



The End